ExamGOAL
PYQ PDF

Free · With answer key

NEET · PHYSICS · CHAPTER-WISE PYQS

Work, Energy and Power

NEET Physics · Previous Year Questions with Answer Key

61	34	2000 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	61 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 61 questions

Q1. RE-NEET 2026

A particle of mass M moves along a horizontal x axis from $x = 0$ to $x = L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L) = 0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is:

Open solution

(A) $\frac{1}{2}$ (B) 3 (C) 1 (D) $\frac{1}{3}$

Q2. NEET 2026

The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is: $(g = 9.8\ \mathrm{m/s^2})$

Open solution

(A) 19.6 W (B) 39.2 W (C) 19.6 kW (D) 39.2 kW

Q3. NEET 2025

The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying brakes, car A stops after 1000 m and car B stops after 1500 m . If F_A and F_B are the forces applied by the brakes on cars A and B respectively, then the ratio of $\frac{F_A}{F_B}$ is

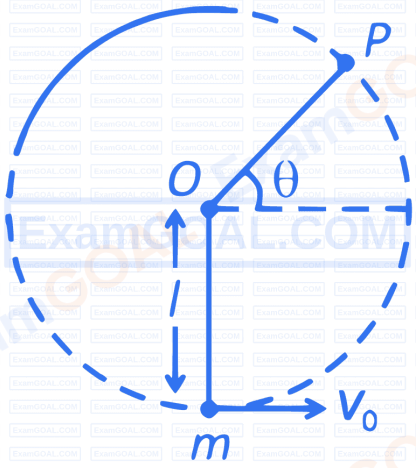
Open solution

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

Q4. NEET 2025

A bob of heavy mass m is suspended by a light string of length l. The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is:

Open solution



(A) $\left(\frac{\cos\theta}{2+3\sin\theta}\right)^{\frac{1}{2}}$ (B) $\left(\frac{\sin\theta}{2+3\sin\theta}\right)^{\frac{1}{2}}$

(C) $(\sin\theta)^{\frac{1}{2}}$ (D) $\left(\frac{1}{2+3\sin\theta}\right)^{\frac{1}{2}}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. NEET 2024 (Re-Examination)

An object moving along horizontal x-direction with kinetic energy $10\ \text{J}$ is displaced through $x = (3\hat{i})\text{m}$ by the force $\vec{F} = (-2\hat{i} + 3\hat{j})\text{N}$. The kinetic energy of the object at the end of the displacement x is

(A) $10\ \text{J}$ (B) $16\ \text{J}$ (C) $4\ \text{J}$ (D) $6\ \text{J}$

Open solution

Q6. NEET 2024 (Re-Examination)

An object falls from a height of $10\ \text{m}$ above the ground. After striking the ground it loses 50% of its kinetic energy. The height upto which the object can rebound from the ground is:

(A) $7.5\ \text{m}$ (B) $10\ \text{m}$ (C) $2.5\ \text{m}$ (D) $5\ \text{m}$

Open solution

Q7. NEET 2024

At any instant of time t, the displacement of any particle is given by $2t - 1$ (SI unit) under the influence of force of $5\ \text{N}$. The value of instantaneous power is (in SI unit):

(A) 10 (B) 5 (C) 7 (D) 6

Open solution

Q8. NEET 2023 Manipur

A particle moves with a velocity $(5\hat{i} - 3\hat{j} + 6\hat{k})\ \text{ms}^{-1}$ horizontally under the action of constant force $(10\hat{\text{i}} + 10\hat{\text{j}} + 20\hat{\text{k}})\text{N}$. The instantaneous power supplied to the particle is :

(A) $200\ \text{W}$ (B) Zero (C) $100\ \text{W}$ (D) $140\ \text{W}$

Open solution

Q9. NEET 2023

The potential energy of a long spring when stretched by $2\ \text{cm}$ is U. If the spring is stretched by $8\ \text{cm}$, potential energy stored in it will be :

(A) 4U (B) 8U (C) 16U (D) 2U

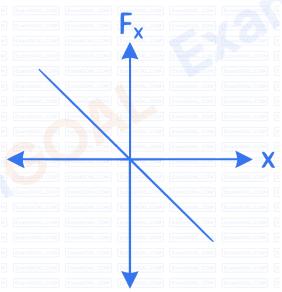
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

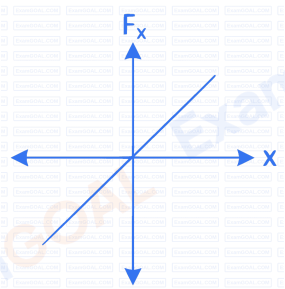
Q10. NEET 2022 Phase 2

The restoring force of a spring with a block attached to the free end of the spring is represented by

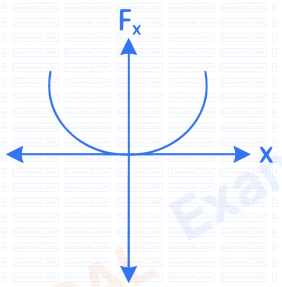
(A)



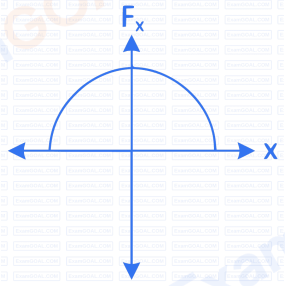
(B)



(C)



(D)



Open solution

Q11. NEET 2022 Phase 1

An electric lift with a maximum load of 2000 kg (lift + passengers) is moving up with a constant speed of 1.5 ms^{-1}. The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is : (g = 10 ms^{-2})

(A) 23000 (B) 20000 (C) 34500 (D) 23500

Open solution

Q12. NEET 2022 Phase 1

The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is

(A) 36×10^7 J (B) 36×10^4 J (C) 36×10^5 J (D) 1×10^5 J

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. NEET 2021

A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively -

Open solution

(A) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$ (B) $\frac{S}{4}, \frac{3gS}{2}$

(C) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$ (D) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$

Q14. NEET 2021

Water falls from a height of 60m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? (g = 10 m/s^2)

Open solution

(A) 7.0 kW (B) 10.2 kW (C) 8.1 kW (D) 12.3 kW

Q15. NEET 2019

A force F = 20 + 10y acts on a particle in y-direction where F is in newton and y in meter. Work done by this force to move the particle from y = 0 to y = 1 m is :

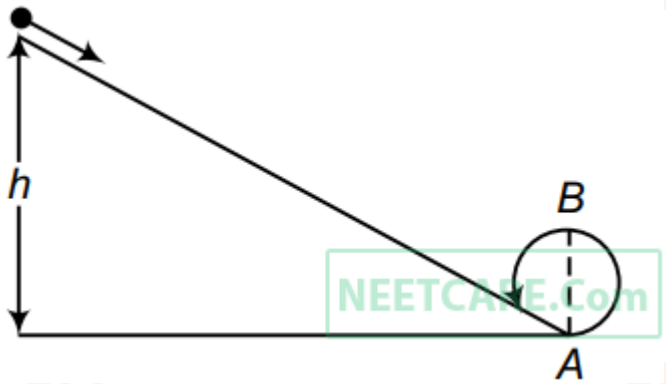
Open solution

(A) 5 J (B) 25 J (C) 20 J (D) 30 J

Q16. NEET 2018

A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter AB = D. The height h is equal to

Open solution



(A) D (B) $\frac{3}{2}D$ (C) $\frac{7}{5}D$ (D) $\frac{5}{4}D$

Q17. NEET 2017

Consider a drop of rain water having mass 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m s^{-1}. Take 'g' constant with a value 10 m s^{-2}. The work done by the (i) gravitational force and the (ii) resistive force of air is

Open solution

(A) (i) 1.25 j (ii) −8.25 J (B) (i) 100 j (ii) 8.75 J

(C) (i) 10 j (ii) −8.75 J (D) (i) −10 j (ii) −8.25 J

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q18. NEET 2016 Phase 2

A particle moves from a point $\left(-2\hat{i}+5\hat{j}\right)$ to $\left(4\hat{j}+3\hat{k}\right)$ when a force of $\left(4\hat{i}+3\hat{j}\right)$ N is applied. How much work has been done by the force ?

(A) 8 J (B) 11 J (C) 5 J (D) 2 J

Open solution

Q19. NEET 2016 Phase 1

A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to 8×10^{-4} J by the end of the second revolution after the beginning of the motion ?

(A) 0.18 m/s^2 (B) 0.2 m/s^2 (C) 0.1 m/s^2 (D) 0.15 m/s^2

Open solution

Q20. NEET 2016 Phase 1

A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F}=\left(2t\hat{i}+3t^2\hat{J}\right)N$, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y axis. What power will be developed by the force at the time t ?

(A) $(2t^3 + 3t^4)$ W (B) $(2t^3 + 3t^5)$ W (C) $(2t^2 + 3t^3)$ W (D) $(2t^2 + 4t^4)$ W

Open solution

Q21. NEET 2016 Phase 1

What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?

(A) $\sqrt{3gR}$ (B) $\sqrt{5gR}$ (C) $\sqrt{gR}$ (D) $\sqrt{2gR}$

Open solution

Q22. AIPMT 2015

Two particles A and B, move with constant velocities $\vec{v_1}$ and $\vec{v_2}$. At the initial moment their position vectors are $\vec{r_1}$ and $\vec{r_2}$ respectively. The condition for particles A and B for their collision is

(A) $\vec{r_1}\times\vec{v_1}=\vec{r_2}\times\vec{v_2}$

(B) $\vec{r_1}-\vec{r_2}=\vec{v_1}-\vec{v_2}$

(C) $\frac{\vec{r_1}-\vec{r_2}}{\left|\vec{r_1}-\vec{r_2}\right|}=\frac{\vec{v_2}-\vec{v_1}}{\left|\vec{v_2}-\vec{v_1}\right|}$

(D) $\vec{r_1}.\vec{v_1}=\vec{r_2}.\vec{v_2}$

Open solution

Q23. AIPMT 2015

The heart of a man pumps 5 litres of blood through the arteries per minute at a pressure of 150 mm of mercury. If the density of mercury be 13.6×10^3 kg/m^3 and g = 10 m/s^2 then the power (in watt) is

(A) 3.0 (B) 1.50 (C) 1.70 (D) 2.35

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q24. AIPMT 2015

A ball is thrown vertically downwards from a height of 20 m with an initial velocity v_0. It collides with the ground, losses 50 percent of its energy in collision and rebounds to the same height. The initial velocity v_0 is

(Take g = 10 m s^{-2})

(A) 28 m s^{-1} (B) 10 m s^{-1} (C) 14 m s^{-1} (D) 20 m s^{-1}

Open solution

Q25. AIPMT 2015

On a frictionless surface, a block of mass M moving at speed v collides elastically with another block of same mass M which is initially at rest. After collision the first block moves at an angle θ to its initial direction and has a speed $\frac{v}{3}$. The second block's speed after the collision is

(A) $\frac{3}{\sqrt{2}}v$ (B) $\frac{\sqrt{3}}{2}v$ (C) $\frac{2\sqrt{2}}{3}v$ (D) $\frac{3}{4}v$

Open solution

Q26. AIPMT 2015 Cancelled Paper

A particle of mass m is driven by a machine that delivers a constant power k watts. If the particle starts from rest the force on the particle at time t is

(A) $\sqrt{2mk}\,t^{-1/2}$ (B) $\frac{1}{2}\sqrt{mk}\,t^{-1/2}$

(C) $\sqrt{\frac{mk}{2}}\,t^{-1/2}$ (D) $\sqrt{mk}\,t^{-1/2}$

Open solution

Q27. AIPMT 2015 Cancelled Paper

A block of mass 10 kg, moving in x direction with a constant speed of 10 m s^{-1}, is subjected to a retarding force F = 0.1x J/m during its travel from x = 20 m to 30 m. Its final KE will be

(A) 275 J (B) 250 J (C) 475 J (D) 450 J

Open solution

Q28. AIPMT 2015 Cancelled Paper

Two similar springs P and Q have spring constants K_P and K_Q, such that $K_P > K_Q$. They are stretched first by the same amount (case a), then by the same force (case b). The work done by the springs W_P and W_Q are related as, in case (a) and case (b) respectively

(A) $W_P > W_Q$; $W_Q > W_P$ (B) $W_P < W_Q$; $W_Q < W_P$

(C) $W_P = W_Q$; $W_P > W_Q$ (D) $W_P = W_Q$; $W_P = W_Q$

Open solution

Q29. NEET 2013 (Karnataka)

A particle with total energy E is moving in a potential energy region U(x). Motion of the particle is restricted to the region when

(A) U(x) < E (B) U(x) = 0 (C) U(x) $\leq$ E (D) U(x) > E

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q30. NEET 2013 (Karnataka)

One coolie takes 1 minute to raise a suitcase through a height of 2 m but the second coolie takes 30 s to raise the same suitcase to the same height. The powers of two coolies are in the ratio

(A) 1 : 3 (B) 2 : 1 (C) 3 : 1 (D) 1 : 2

Open solution

Q31. NEET 2013

A uniform force of $\left(3\hat{i}+\hat{j}\right)$ newton acts on a particle of mass 2 kg. Hence the particle is displaced from position $\left(2\hat{i}+\hat{k}\right)$ metre to position $\left(4\hat{i}+3\hat{j}-\hat{k}\right)$ metre. The work done by the force on the particle is

(A) 13 J (B) 15 J (C) 9 J (D) 6 J

Open solution

Q32. AIPMT 2012 Mains

A car of mass m starts from rest and accelerates so that the instantaneous power delivered to the car has a constant magnitude P_0. The instantaneous velocity of this car is proportional to

(A) t^2P_0 (B) $t^{1/2}$ (C) $t^{-1/2}$ (D) $\frac{t}{\sqrt{m}}$

Open solution

Q33. AIPMT 2012 Prelims

The potential energy of a particle in a force field is $U=\frac{A}{r^2}-\frac{B}{r}$ where A and B are positive constants and r is the distance of particle from the centre of the field. For stable equilibrium, the distance of the particle is

(A) $\frac{B}{2A}$ (B) $\frac{2A}{B}$ (C) $\frac{A}{B}$ (D) $\frac{B}{A}$

Open solution

Q34. AIPMT 2012 Prelims

A solid cylinder of mass 3 kg is rolling on a horizontal surface with velocity 4 m s^{-1}. It collides with a horizontal spring of force constant 200 N m^{-1}. The maximum compression produced in the spring will be

(A) 0.5 m (B) 0.6 m (C) 0.7 m (D) 0.2 m

Open solution

Q35. AIPMT 2011 Mains

A mass m moving horizontally (along the x-axis) with velocity v collides and sticks to a mass of 3m moving vertically upwards (along the y-axis) with velocity $2v$. The final velocity of the combination is

(A) $\frac{3}{2}v\hat{i}+\frac{1}{4}v\hat{j}$ (B) $\frac{1}{4}v\hat{i}+\frac{3}{2}v\hat{j}$

(C) $\frac{1}{3}v\hat{i}+\frac{2}{3}v\hat{j}$ (D) $\frac{2}{3}v\hat{i}+\frac{1}{3}v\hat{j}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q36. AIPMT 2011 Prelims

The potential energy of a system increases if work is done

(A) upon the system by a nonconservative force

(B) by the system against a conservative force.

(C) by the system against a nonconservative force.

(D) upon the system by a conservative force.

Open solution

Q37. AIPMT 2011 Prelims

A body projected vertically from the earth reaches a height equal to earth's radius before returning to the earth. The power exerted by the gravitational force is greatest

(A) at the highest position of the body

(B) at the instant just before the body hits the earth.

(C) it remains constant all through.

(D) at the instant just after the body is projected.

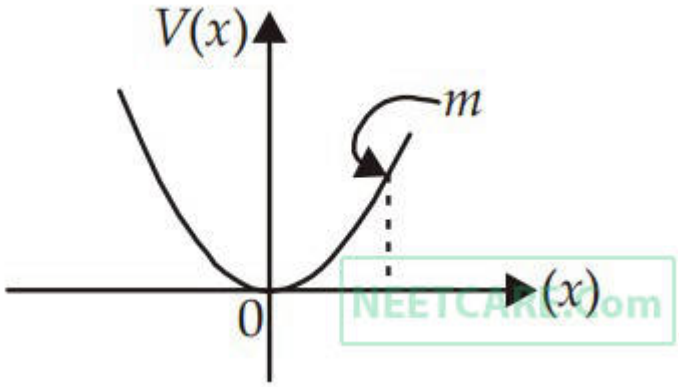
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

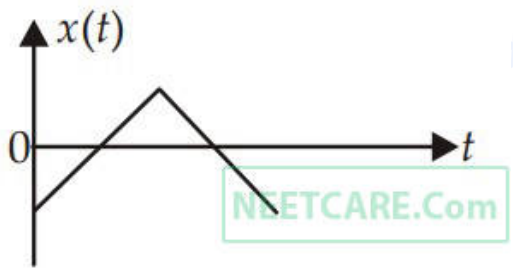
Q38. AIPMT 2011 Prelims

A particle of mass m is released from rest and follows a parabolic path as shown. Assuming that the displacement of the mass from the origin is small, which graph correctly depicts the position of the particle as a function of time ?

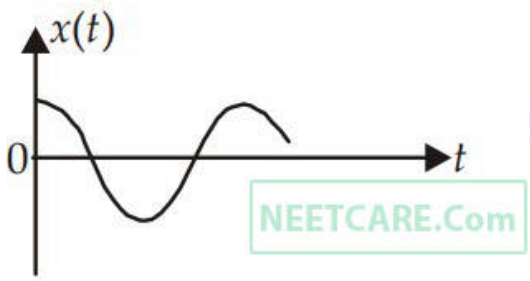
Open solution



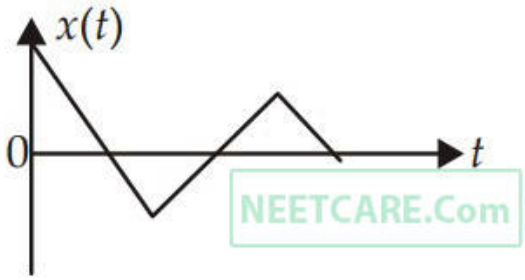
(A)



(B)

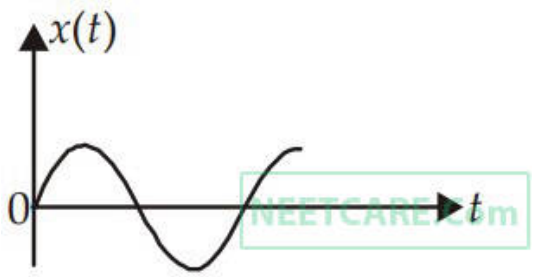


(C)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

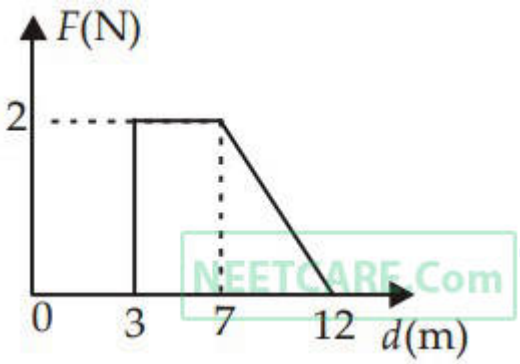
(D)



Q39. AIPMT 2011 Prelims

Force F on a particle moving in a straight line varies with distance d as shown in figure.

Open solution



The work done on the particle during its displacement of 12 m is

(A) 18 J (B) 21 J (C) 26 J (D) 13 J

Q40. AIPMT 2010 Mains

A particle of mass M, starting from rest, undergoes uniform acceleration. If the speed acquired in time T is V, the power delivered to the particle is

Open solution

(A) $\frac{MV^2}{T}$ (B) $\frac{1}{2}\frac{MV^2}{T^2}$

(C) $\frac{MV^2}{T^2}$ (D) $\frac{1}{2}\frac{MV^2}{T}$

Q41. AIPMT 2010 Prelims

A ball moving with velocity 2 m/s collides head on with another stationary ball of double the mass. If the coefficient of restitution is 0.5, then their velocities (in m/s) after collision will be

Open solution

(A) 0, 1 (B) 1, 1 (C) 1, 0.5 (D) 0, 2

Q42. AIPMT 2010 Prelims

An engine pumps water through a hose pipe. Water passes through the pipe and leaves it with a velocity of 2 m/s. The mass per unit length of water in the pipe is 100 kg/m. What is the power of the engine?

Open solution

(A) 400 W (B) 200 W (C) 100 W (D) 800 W

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q43. AIPMT 2009

A block of mass M is attached to the lower end of a vertical spring. The spring is hung from a ceiling and has force constant value k. The mass is released from rest with the spring initially unstretched. The maximum extension produced in the length of the spring will be

Open solution

(A) 2Mg/k (B) 4Mg/k (C) Mg/2k (D) Mg/k

Q44. AIPMT 2009

A body of mass 1 kg is thrown upwards with a velocity 20 m/s. It momentarily comes to rest after attaining a height of 18 m. How much energy is lost due to air friction? (g = 10 m/s^2)

Open solution

(A) 30 J (B) 40 J (C) 10 J (D) 20 J

Q45. AIPMT 2009

An engine pumps water continuously through a hose. Water leaves the hose with a velocity v and m is the mass per unit length of the water jet. What is the rate at which kinetic energy is imparted to water?

Open solution

(A) mv^3 (B) $\frac{1}{2}mv^2$ (C) $\frac{1}{2}m^2v^2$ (D) $\frac{1}{2}mv^3$

Q46. AIPMT 2008

Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional forces are 10% of energy. How much power is generated by the turbine ? (g = 10 m/s^2)

Open solution

(A) 12.3 kW (B) 7.0 kW (C) 8.1 kW (D) 10.2 kW

Q47. AIPMT 2007

A vertical spring with force constant k is fixed on a table. A ball of mass m at a height h above the free upper end of the spring falls vertically on the spring so that the spring is compressed by a distance d. The net work done in the process is

Open solution

(A) $mg(h + d) - \frac{1}{2}kd^2$ (B) $mg(h - d) - \frac{1}{2}kd^2$

(C) $mg(h - d) + \frac{1}{2}kd^2$ (D) $mg(h + d) + \frac{1}{2}kd^2$

Q48. AIPMT 2006

300 J of work is done in sliding a 2 kg block up an inclined plane of height 10 m. Work done against friction is (Take g = 10 m/s^2)

Open solution

(A) 1000 J (B) 200 J (C) 100 J (D) zero

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q49. AIPMT 2006

The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm the potential energy stored in it is

(A) U/4 (B) 4U (C) 8U (D) 16U

Open solution

Q50. AIPMT 2006

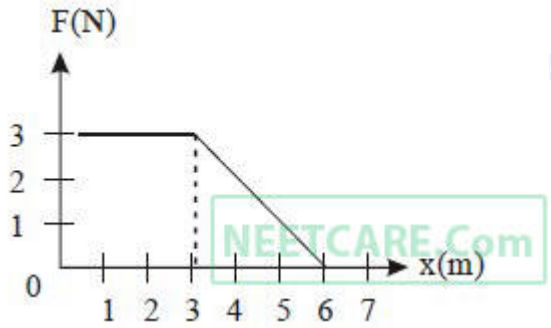
A body of mass 3 kg is under a constant force which causes a displacement s in metres in it, given by the relation s = $\frac{1}{3}t^2$, where t is in seconds. Work done by the force in 2 seconds is

(A) $\frac{19}{5}J$ (B) $\frac{5}{19}J$ (C) $\frac{3}{8}J$ (D) $\frac{8}{3}J$

Open solution

Q51. AIPMT 2005

A force F acting



on an object varies with distance x as shown here. The force is in N and x in m. The work done by the force in moving the object from x = 0 to x = 6 m is

(A) 18.0 J (B) 13.5 J (C) 9.0 J (D) 4.5 J

Open solution

Q52. AIPMT 2004

A particle of mass m_1 is moving with a velocity v_1 and another particle of mass m_2 is moving with a velocity v_2. Both of them have the same momentum but their different kinetic energies are E_1 and E_2 respectively. If $m_1 > m_2$ then :

(A) $E_1 < E_2$ (B) $\frac{E_1}{E_2} = \frac{m_1}{m_2}$

(C) $E_1 > E_2$ (D) $E_1 = E_2$

Open solution

Q53. AIPMT 2004

A ball of mass 2 kg and another of mass 4 kg are dropped together from a 60 feet tall building. After a fall of 30 feet each towards earth, their respective kinetic energies will be in the ratio of

(A) $\sqrt{2}:1$ (B) 1 : 4 (C) 1 : 2 (D) $1:\sqrt{2}$

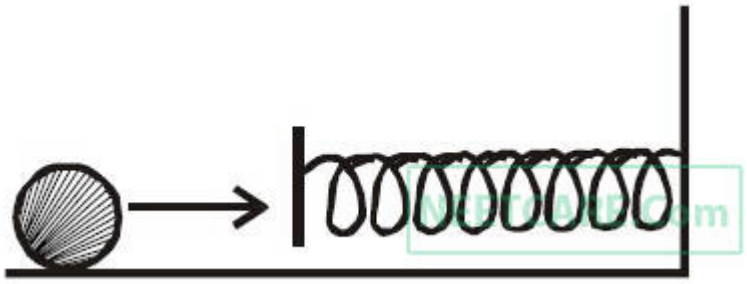
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q54. AIPMT 2004

A mass of 0.5 kg moving with a speed of 1.5 m/s on horizontal smooth surface, collides with a nearly weightless spring of force constant k = 50 N/m. The maximum compression of the spring would be

Open solution



(A) 0.15 m (B) 0.12 m (C) 1.5 m (D) 0.5 m

Q55. AIPMT 2003

A particle moves along a circle of radius $\left(\frac{20}{\pi}\right)$ m with constant tangential acceleration. If the velocity of the particle is 80 m/s at the end of the second revolution after motion has begun, the tangential acceleration is

Open solution

(A) 40 m/s^2 (B) 640π m/s^2 (C) 160π m/s^2 (D) 40π m/s^2

Q56. AIPMT 2003

When a long spring is stretched by 2 cm, its potential energy is U. If the spring is stretched by 10 cm, the potential energy stored in it will be

Open solution

(A) U/5 (B) 5U (C) 10U (D) 25U

Q57. AIPMT 2001

A child is sitting on a swing. Its minimum and maximum heights from the ground 0.75 m and 2 m respectively, its maximum speed will be

Open solution

(A) 10 m/s (B) 5 m/s (C) 8 m/s (D) 15 m/s

Q58. AIPMT 2001

Two springs A and B having spring constant K_A and K_B ($K_A = 2K_B$) are stretched by applying force of equal magnitude. If energy stored in spring A is E_A then energy stored in B will be

Open solution

(A) $2E_A$ (B) $E_A/4$ (C) $E_A/2$ (D) $4E_A$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q59. AIPMT 2001

A particle is projected making an angle of 45° with horizontal having kinetic energy K. The kinetic energy at highest point will be

(A) $\frac{K}{\sqrt{2}}$ (B) $\frac{K}{2}$ (C) 2K (D) K

Open solution

Q60. AIPMT 2000

If $\overrightarrow{F} = \left(60\hat{i} + 15\hat{j} - 3\hat{k}\right)$ N and $\overrightarrow{v} = \left(2\hat{i} - 4\hat{j} + 5\hat{k}\right)$ m/s, then instantaneous power is

(A) 195 watt (B) 45 watt (C) 75 watt (D) 100 watt.

Open solution

Q61. AIPMT 2000

A mass of 1 kg is thrown up with a velocity of 100 m/s. After 5 seconds, it explodes into two parts. One part of mass 400 g comes down with a velocity 25 m/s. The velocity of other part is (Take g = 10 ms^{-2})

(A) 40 m/s (B) 50 m/s (C) 100 m/s (D) 60 m/s

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

1. **A**	2. **C**	3. **D**	4. **B**	5. **C**
6. **D**	7. **A**	8. **D**	9. **C**	10. **A**
11. **C**	12. **A**	13. **A**	14. **C**	15. **B**
16. **D**	17. **C**	18. **C**	19. **C**	20. **B**
21. **B**	22. **C**	23. **C**	24. **D**	25. **C**
26. **C**	27. **C**	28. **A**	29. **C**	30. **D**
31. **C**	32. **B**	33. **B**	34. **B**	35. **B**
36. **B**	37. **B**	38. **B**	39. **D**	40. **D**
41. **A**	42. **D**	43. **A**	44. **D**	45. **D**
46. **C**	47. **A**	48. **C**	49. **D**	50. **D**
51. **B**	52. **A**	53. **C**	54. **A**	55. **A**
56. **D**	57. **B**	58. **A**	59. **B**	60. **B**
61. **C**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs